

Week1 Homework

By Deng Yufan without AI.

Problem 5 (P108 T6)

We have a connected graph $G = (V, E)$, and a specific vertex $u \in V$. Suppose we compute a depth-first search tree rooted at u , and obtain a tree T that includes all nodes of G . Suppose we then compute a breadth-first search tree rooted at u , and obtain the same tree T . Prove that $G = T$. (In other words, if T is both a depth-first search tree and a breadth-first search tree rooted at u , then G cannot contain any edges that do not belong to T .)

If G is directed, then this statement is false since a 3-vertex ring breaks this statement. So we suppose G is undirected.

We know that in DFS tree there is no crossing edge (i.e. 横叉边), and in BFS tree there is no backing edge (i.e. 返祖边). So since T is both DFS tree and BFS tree, there is neither crossing edge nor backing edge in T , that is $T = G$.

Problem 6

We are given $2n$ balls and n bins. Each ball is thrown into a random bin (each bin is chosen with probability $1/n$). Compute the following limit

$$\lim_{n \rightarrow +\infty} \mathbb{E}\left[\frac{\text{the number of empty bins}}{n}\right].$$

We have

$$\mathbb{E}\left[\frac{\text{the number of empty bins}}{n}\right] = \Pr(i\text{-th bin is empty}) = \left(\frac{n-1}{n}\right)^{2n}.$$

Then

$$\lim_{n \rightarrow +\infty} \left(\frac{n-1}{n}\right)^{2n} = \left[\lim_{n \rightarrow +\infty} \left(\frac{n-1}{n}\right)^n\right]^2 = \frac{1}{e^2}.$$

Problem 7

For two women w and w' , we write $w <_m w'$ to denote that w is worse than w' in the preference list of man m . Given two stable matchings f and f' (You can easily construct an example in which there are multiple stable matchings), define a mapping $g = f \vee f'$ as follows:

- for each man m , assign him more preferred partner

$$g(m) = f(m) \text{ if } f(m) \geq_m f'(m)$$

$$g(m) = f'(m) \text{ if } f'(m) >_m f(m)$$

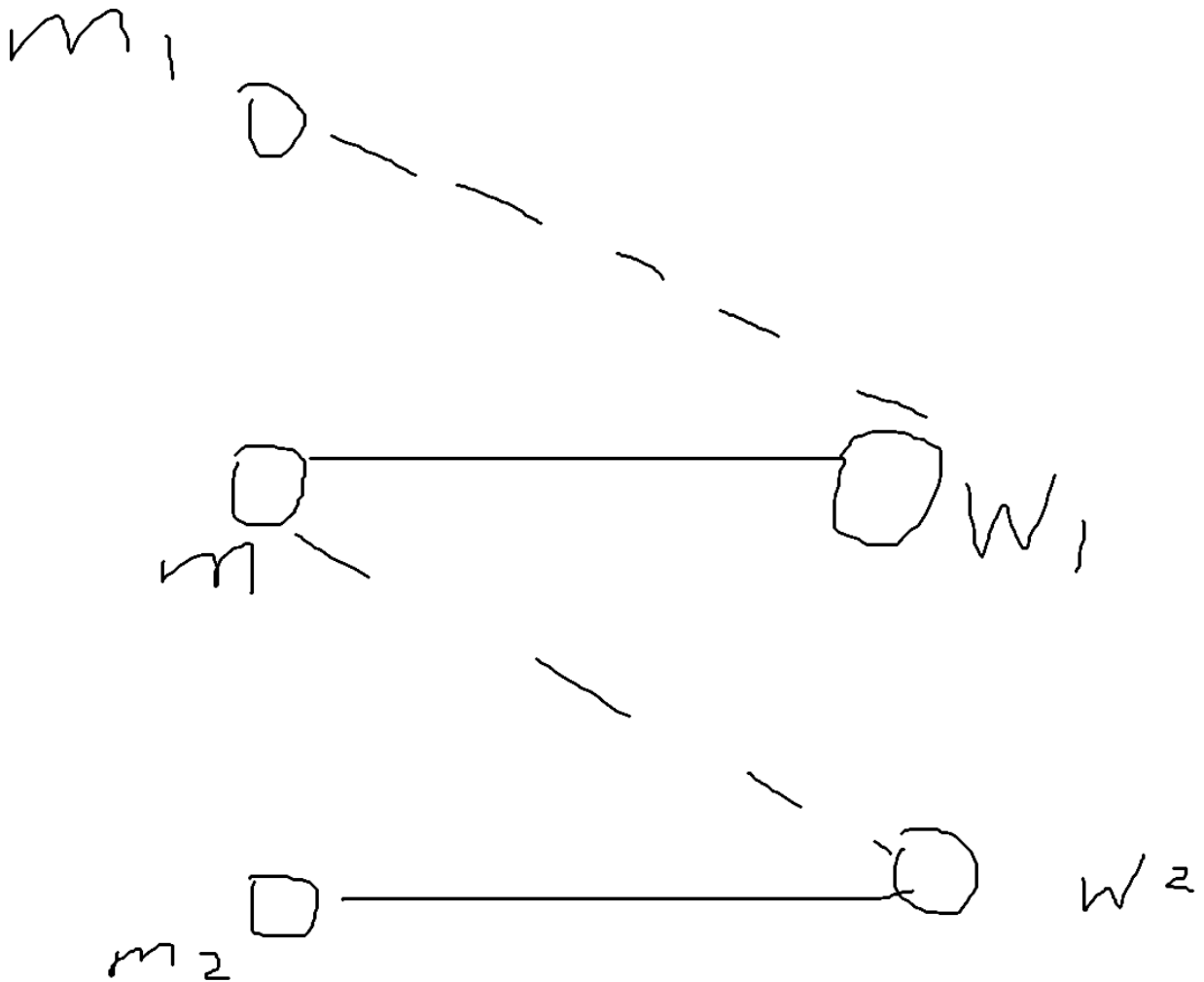
- for each woman w , assign her less preferred partner

$$g(w) = f(w) \text{ if } f(w) \leq_w f'(w)$$

$$g(w) = f'(w) \text{ if } f'(w) <_w f(w)$$

Show that if both f and f' are stable matchings, so is g . (note: We can similarly define $f \wedge f'$. Then, all stable matchings form a distributive lattice, an abstract yet very popular object studied in combinatorics.)

We first prove g is a matching. For a man m , let $w_1 = f(m)$, $w_2 = f'(m)$, $m_1 = f'(w_1)$, $m_2 = f(w_2)$, as the figure.



In stable matching f , we have (m, w_2) is not a unstable match, that is $w_1 >_m w_2 \vee m <_{w_2} m_2$ holds. Similarly, since (m, w_1) is not a unstable match in f' , we have $w_1 <_m w_2 \vee m <_{w_1} m_1$ holds.

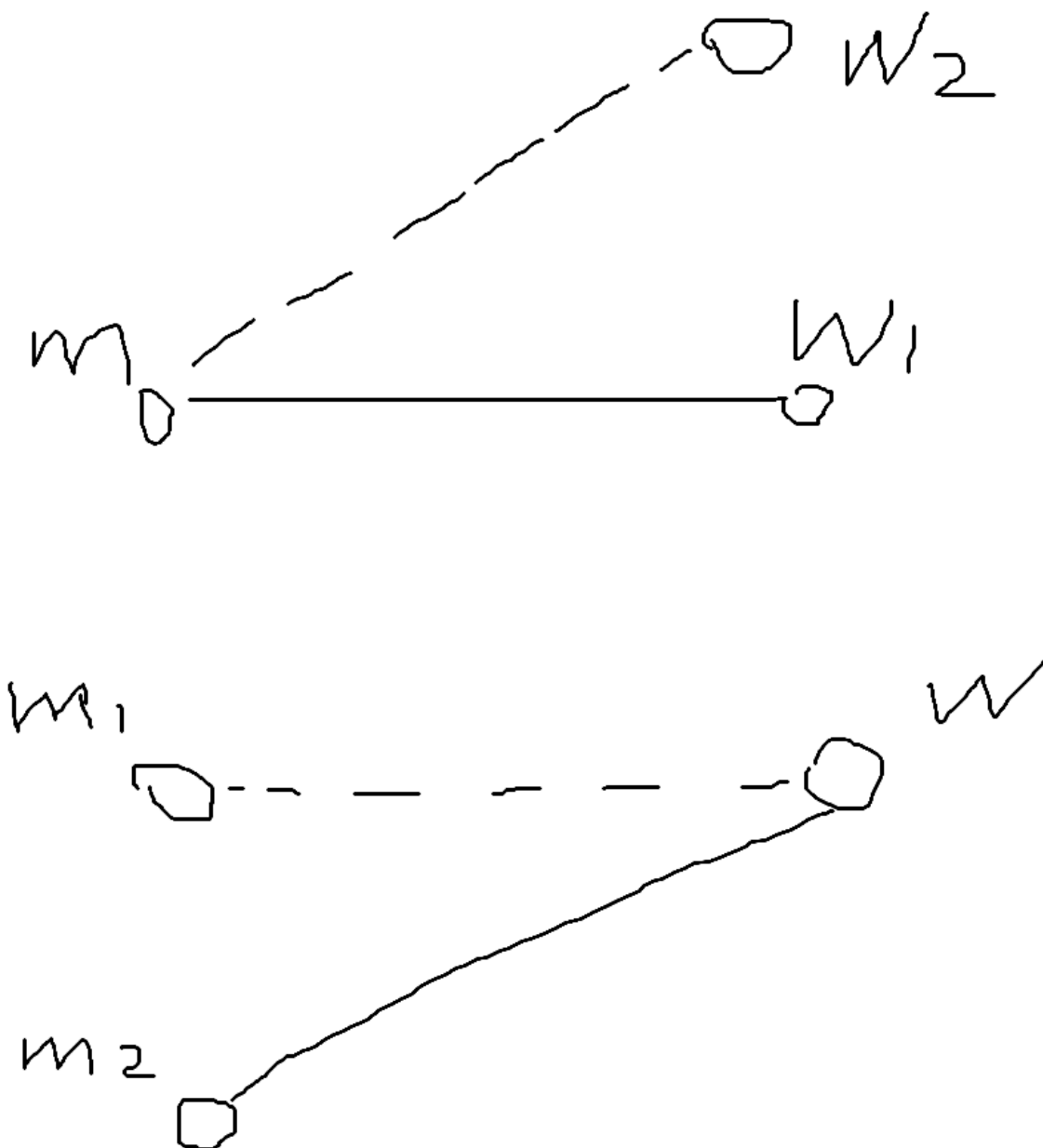
Suppose $w_1 >_m w_2$, or say $g(m) = w_1$, then $w_1 <_m w_2 \vee m <_{w_1} m_1$ deduces $m <_{w_1} m_1$, that is $g(w_1) = m$. If $w_1 <_m w_2$, similarly, we can deduce that $g(m) = w_2, g(w_2) = m$.

Now we claim that for each man m , we have $g(g(m)) = m$, thus g is bijection, or say g is a valid matching.

Suppose (m, w) is a unstable match in g . Let $w_1 = g(m), m_1 = g(w)$, so we have $w >_m w_1$ and $m >_w m_1$.

If $f(m) = w_1$ and $f(w) = m_1$, then (m, w) is a unstable match in f , contradiction; if $f'(m) = w_1$ and $f'(w) = m_1$, then (m, w) is a unstable match in f' , contradiction. So we have $f(m) = w_1 \neq f'(m)$ and $f'(w) = m_1 \neq f(w)$ (or $f'(m) = w_1 \neq f(m)$ and $f(w) = m_1 \neq f'(w)$, but is similar because f and f' can swap).

Let $w_2 = f'(m), m_2 = f(w)$, as the figure.



Since $g(m) = f(m) = w_1$ and $g(m) \neq f'(m) = w_2$, we have $w_1 >_m w_2$.

Since (m, w) is not a unstable match in f' , and $m \succ_w m_1$, we have $w_2 \succ_m w$.

And we already have $w \succ_m w_1$. That is $w_1 \succ_m w_2 \succ_m w \succ_m w_1$, contradiction. So (m, w) is not a unstable match, thus g is stable.